2017 F=ma Exam: Problem 1

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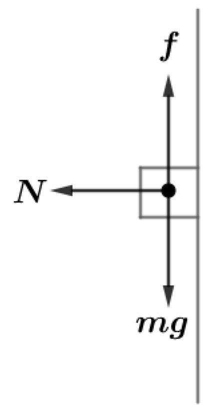

Consider the forces on the motorcycle. The normal force provides the centripetal acceleration so we have

$$N = \frac{ms^2}{R}$$

where s is the speed of the motorcycle and R is the radius of the circular room. For the motorcycle not to slip downwards,

$$f = mg$$

Since $f \leq \mu N$,

$$mg \leq \frac{\mu m s^2}{R}$$

Thus

$$\mu \geq \frac{gR}{s^2}$$

$$\mu_{\text{min}} \propto s^{-2}$$

so the answer is $\boxed{\text{D}}$.